Name: Savitha R

Reg.no: 192421368

19 Implementation of Naive Bayes classification for Bank Loan prediction

Program:

```
# Naive Bayes Classification
# Bank Loan Prediction

from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix

# Dataset
# Features:
# [Age, Income, Loan Amount, Credit Score]

X = [
    [25, 25000, 5000, 650],
    [30, 35000, 7000, 680],
    [35, 45000, 10000, 720],
    [40, 50000, 12000, 750],
    [45, 60000, 15000, 780],
    [50, 70000, 18000, 800],
    [28, 30000, 6000, 660],
    [32, 40000, 8000, 700],
    [38, 48000, 11000, 730],
    [42, 55000, 13000, 760],
    [48, 65000, 16000, 790],
    [52, 75000, 20000, 820],
```

```
    [24, 22000, 4500, 620],
    [27, 28000, 5500, 640],
    [36, 42000, 9000, 710],
    [44, 58000, 14000, 770]
]

# Loan Approval
# 0 = Not Approved
# 1 = Approved

y = [
    0, 0, 1, 1,
    1, 1, 0, 1,
    1, 1, 1, 1,
    0, 0, 1, 1
]

# Split the dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.3,
    random_state=42
)

# Create Naive Bayes classifier
model = GaussianNB()

# Train the model
model.fit(X_train, y_train)
```

```
# Predict test data
y_pred = model.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

# Calculate confusion matrix
cm = confusion_matrix(y_test, y_pred)

# Display results
print("========================================")
print("   BANK LOAN PREDICTION")
print("   USING NAIVE BAYES")
print("========================================")

print("\nActual Values:")
print(y_test)

print("\nPredicted Values:")
print(y_pred)

print("\nAccuracy:")
print(accuracy)

print("\nAccuracy Percentage:")
print(accuracy * 100, "%")

print("\nConfusion Matrix:")
print(cm)
```

```
# Predict a new customer
new_customer = [[33, 42000, 9000, 710]]

prediction = model.predict(new_customer)

print("\nNew Customer Details:")
print("Age:", new_customer[0][0])
print("Income:", new_customer[0][1])
print("Loan Amount:", new_customer[0][2])
print("Credit Score:", new_customer[0][3])

print("\nLoan Prediction:")

if prediction[0] == 1:
    print("Loan Approved")
else:
    print("Loan Not Approved")
```
output:

```
======================== RESTART: E:/Machine Learning/exp 19.py =
=====================================
   BANK LOAN PREDICTION
   USING NAIVE BAYES
=====================================

Actual Values:
[0, 0, 1, 1, 0]

Predicted Values:
[0 0 1 1 0]

Accuracy:
1.0

Accuracy Percentage:
100.0 %

Confusion Matrix:
[[3 0]
 [0 2]]

New Customer Details:
Age: 33
Income: 42000
Loan Amount: 9000
```


```
Accuracy Percentage:
100.0 %

Confusion Matrix:
[[3 0]
 [0 2]]

New Customer Details:
Age: 33
Income: 42000
Loan Amount: 9000
```


```
Credit Score: 710

Loan Prediction:
Loan Approved
```